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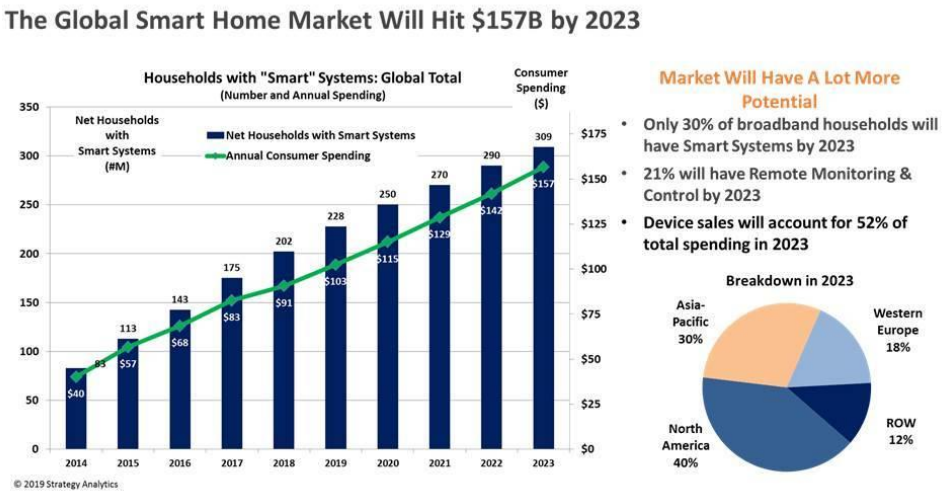
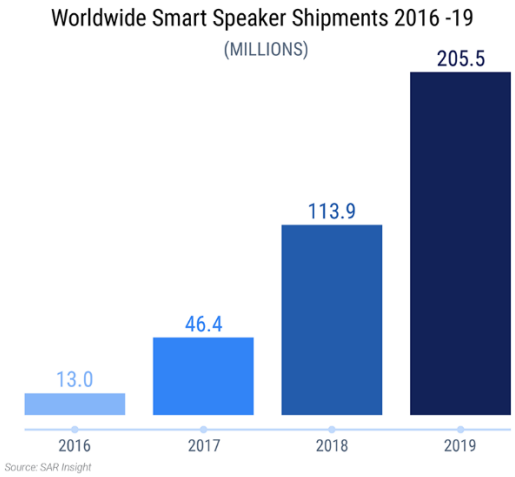
Tohoku University 2022

SMART HOME EXPERIMENTS





https://www.businesswire.com/news/home/20190925005582/en/Strategy-Analytics-Global-Smart-Home-Market-to-Surpass-100-Billion-in-2019

https://voicebot.ai/2019/09/24/smart-speaker-sales-to-rise-35-globally-in-2019-to-92-million-units-15-million-in-china-growth-slows/

https://www.forbes.com/sites/ericmack/2014/02/19/elon-musk-tesla-will-be-first-with-autonomous-driving-admits-to-apple-meeting/?sh=239bc0d53724

2014

My interest of developing smart IOT products started around 2014 after witnessing TESLAs success in producing their first real world practical version of a self-driving autonomous car. Communications technology development in the construction sector is slower than in the manufacturing sector. Therefore, the development of autonomous housing will soon be practical in the market.

2015

Early investigations cantered around wired home components. A collection of lights, speakers, chargers can provide housing items such as tables, chairs and beds with electrical energy to perform various task that was traditionally not expected of such household furniture. However, the components were still rather expensive and the products created would not be at an interesting price point for the market.

2018

Smart household fittings such as switches and lights started to be available in the market with varying degree of effectiveness and practicality. Early smart switches required a neutral wire connection making it tricky and costly for installation in renovation projects.

2019

Development of Smart home products is picking up globally. China produced components begin to rival European products with decent quality at a fraction of the cost. Early players in the local Malaysian market start to bring in products in bulk from China providing warranties for the products sold locally. More products come with wireless capability, eliminating the need for costly wiring works.

2020

Products available still does not have 100% performance reliability. Early adopters of the products are Smart home hobbyist and tech savvy individuals. Wired smart homes still provide high end clients with better peace of mind and reliability. However future adaptability is very limited for wired smart home systems.

In order to perform real world test with newer, more affordable smart home components that was unproven, I offered free smart home upgrades to my clients. The free aspect of the experiment makes it more acceptable for casual smart home enthusiast. It also reduces the liability inherent in a cheaper and less reliable wireless smart home system.

A small percentage of the project cost was allocated to experiment with smart home products available in the Malaysian market.

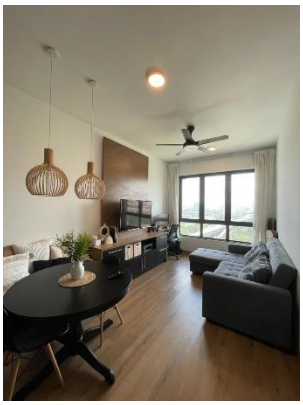
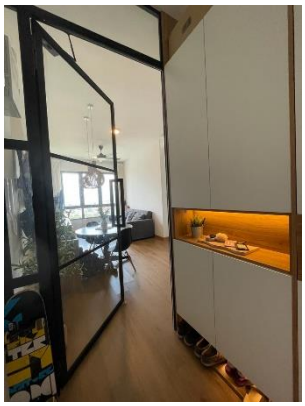
TEST INSTALLATION, 2020

1st Condominium, Bukit Jalil

The first project was a trial of the effectiveness of the Google Home smart speaker to control an IR sensor within a single space. This enabled the control of the TV , Fan and Aircon remote by voice.

2nd Bungalow, Sungai Penchala

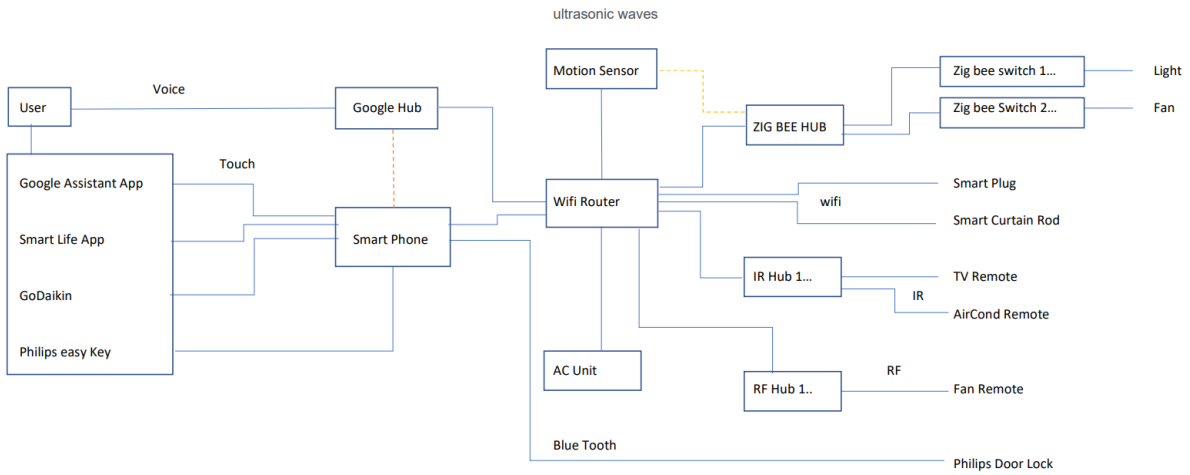
The next project was trying to link the voice control system with the wired lighting and fan switch. Motion sensors were used here. The space was also larger with two Fans, and two Airconditioning units.
To avoid any issue with the smart home system, we partnered with https://www.explosoft.com.my/ to provide the parts warranty and installation support.
The products were offered at a good price, however the Manufacturer limited the components adaptability with other 3rd party products.

3rd Condominium, Menjalara

This project provided more flexibility to experiment as it was a personal project. The basic methods needed to link the smart home was established in the earlier projects. However, this time products were sourced openly to enable long term adaptability. Additional smart home components could be added into the system when it became available. The zigbee wireless network was used to enable more components to join in the system without crowding the wifi network. Multiple of each component were used to cater to multiple spaces. However, vampire load drives up the electrical bill cost as opposed to a non-connected home.

SMART HOME NETWORK

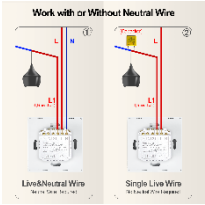


Wifi Router: Main component enabling wifi connection in a housing unit. Distributes the incoming fibre internet into a wireless network.

Google Smart Speaker: Microphone and speaker component connected to googles cloud AI system. Smart Speakers from other manufacturers provides similar connection services with varying capability. Allows voice control, web connectivity and phone compatibility of various smart home systems.

Zigbee Hub: Provides an alternative wireless network with more connection capacity without overcrowding and over depending on the Wifi network. Collects information from various zigbee components and provides a single point of contact with the Wifi router. The zigbee is a local wireless network and is uninterrupted in the event that the Wifi network is interrupted.

Zigbee Smart Switch: In addition to its function as a traditional on/off switch. The smart switch provides information to the network if it is in the on or off state. The network can then send



instruction to the switch wirelessly to change its state from on or off or vice versa. Earlier generation of smart switches require a neutral wire to function.

IR / RF Remote: This device copies Infrared or Radio Frequencies from other standard household remotes. The digitalised remote can then be controlled through the smart home network via this device.

Sensors: Various sensors can be plugged into the smart home network to provide information to the system.

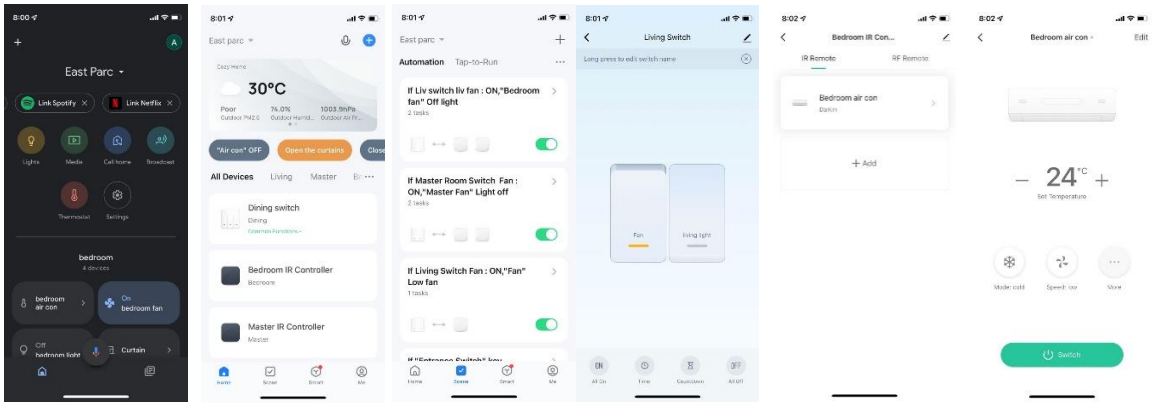
This motion sensor emits ultrasonic sound waves and detects movement when the sound wave is disrupted.

Smart Curtain: This plugged in electric smart curtain was chosen over battery operated smart curtains because motor movement requires higher energy. Battery system will wear out fast thus making it impractical for its intended use. The motor receives wireless information from the network regarding the intended position of the curtain.

SMART HOME INTERFACE

The smart home network is primarily controlled by touch through a smart phone device or by voice through the cloud network enabled Smart Speaker.

Smart Phone interface



The main smart phone device interface is provided by 'Google Home app'. Further refinement can be controlled via 'Smart Life App' which has a link function with 'Google Home app'. Certain degree of automation is possible through the app.

Smart Speaker interface

Say "Okay Google" followed by commands as below:

No	Room/Area	Home Appliance	Command	Name
1	Entrance	Entrance Light	Swich on/switch off	"Doorway"
		Key Cabinet Light	Swich on/switch off	"Key Cabinet"
		Motion Sensor	-	-
2	Kitchen	Stove main switch	Swich on/switch off	"Stove socket" or "Outlet"
		Kitchen Light	Swich on/switch off	"Kitchen Light"
4	Dining Area	Dining hanging light	Swich on/switch off	"Dining Light"
5	Living Area	Smart TV Model : LG 49UM7290PTD	Swich on/switch off	"TV" or "Living TV"
		Fan	Swich on/switch off	"Living Fan"
		Living light	Swich on/switch off	"Living Light"
		Curtain	Open / Close	"Living Curtain"
		Light behind TV panel	Swich on/switch off	"TV light"
		Corridor Light	Swich on/switch off	"Corridor light)
		Aircond	Swich on/switch off	"Living Air"
6	Bedroom 2	Light	Swich on/switch off	"Bedroom Light"
		Fan	Swich on/switch off	"Bedroom Fan"
		Curtain	Open / Close	"Bedroom Curtain"
7	Master Bedroom	TV	Swich on/switch off	"TV" or "Master TV"
		Light	Swich on/switch off	"Master Light"
		Aircond	Swich on/switch off	"Master Aircond"
		Curtain	Open / Close	"Master Curtain"
		Fan	Swich on/switch off	"Master Fan"

These are some of the standard instructions possible via the smart speaker device. Googles speech recognition AI allows for various other words to be used besides the one stated above.

LIMITATIONS

Network limit.

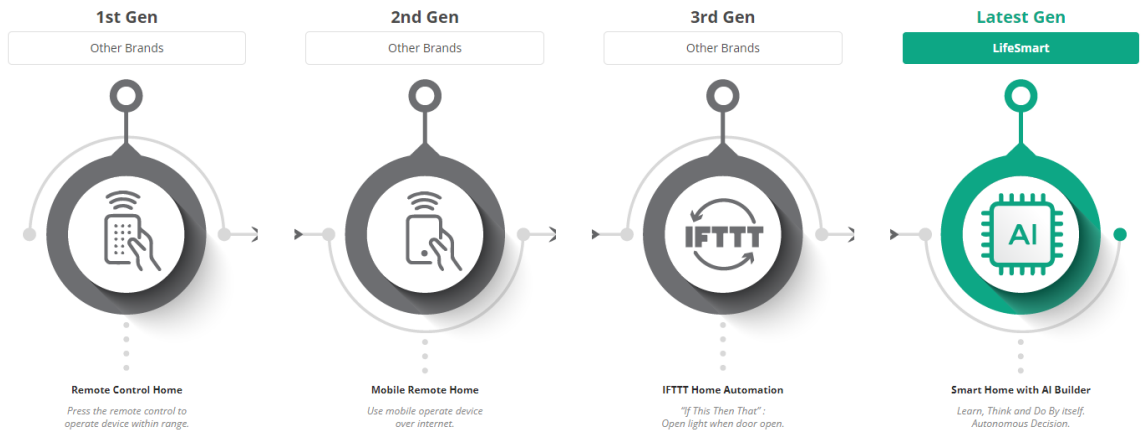
One limitation in the system is that the network begins to get crowded when more components are installed in the network causing signal delays or mixed signals. A lot of the components still rely heavily on the wifi network to connect to the smart home system. Wifi networks are often burdened by data load from regular internet usage.

Each wireless network has also a range limit and penetration limit when implemented in the built environment.

The wireless network is also less stable. Certain components can be missing from the system for short periods of time causing the system to get confused on how to execute the desired command.

Intelligence limit.

Another issue with this smart home network system is that the system is not truly smart. Although the devices are connected to the web and is able to be controlled remotely, almost all of the instructions has to be given into the network by the user.

Any automation programmed into the current system falls under the IFTTT information model. IFTTT stands for “If this than that”. It is a set of precise instructions given to the network to trigger certain settings. This system lacks any real Artificial Intelligence to predict and learn from home usage trends. The IFTTT model is also limited due to the complex situations of household electrical usage patterns.

The next generation of smart houses (as suggested by https://lifesmart.com.my/) with cloud based Machine learning AI systems will be able to learn, predict and execute the homes controls more appropriately. One suggestion to achieve this is through Microsoft AI builder.